

Research Proposal Components

Federated Traffic Sign Recognition under Non-IID Label Distributions: A Reproducible Study of FedAvg and FedProx

Student Name: Ma He

Faculty supervisor: Song Linqi

Institution: City University of Hong Kong

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Abstract

Federated learning enables multiple clients to train a shared model without transferring raw data to a central server, but heterogeneous client data may reduce global performance. This project investigates how label-distribution skew affects federated traffic-sign recognition and whether FedProx can mitigate the resulting degradation. Experiments use the 43-class German Traffic Sign Recognition Benchmark with five simulated clients. Centralised, Local-only, and IID FedAvg baselines have been established, followed by fixed Dirichlet partitions with $\alpha=10, 1.0, 0.5$, and 0.1 . Preliminary results show that FedAvg remains stable under mild and moderate skew but deteriorates substantially at $\alpha=0.1$, where test accuracy falls to 71.49% and several classes receive zero accuracy. In strictly matched experiments, increasing the FedProx proximal strength from $\mu=0$ to $\mu=0.1$ improves test accuracy from 77.51% to 84.52% and macro-class accuracy from 73.60% to 80.79% at $\alpha=0.1$. The proposed study will consolidate these findings through reproducible overall and class-level analysis, identify the robustness boundary of FedAvg, and evaluate the extent to which proximal regularisation provides meaningful but incomplete mitigation under strong label heterogeneity.

1 Introduction and Significance

Traffic sign recognition is an important component of intelligent transportation systems and automated driving. Vehicles must accurately identify speed-limit, warning, prohibition, and mandatory signs in order to support driving decisions and improve road safety. Training an effective recognition model generally requires large image collections obtained from different vehicles, geographical areas, and road environments. However, transferring all raw images to a central server may create concerns relating to privacy, data ownership, storage capacity, and communication cost.

Federated learning offers a practical approach to collaborative model training with distributed data. Instead of uploading raw images, each client trains a model locally and sends model parameters or updates to a central server. The server aggregates these updates to produce a shared global model, which is then redistributed for the next training round. This approach reduces the need for centralised data sharing, but its effectiveness can be strongly influenced by differences among client data distributions.

In realistic vehicular environments, clients are unlikely to observe identical traffic sign categories or frequencies. One vehicle may encounter mainly speed-limit signs, whereas another may collect more warning or prohibition signs. This form of label-distribution skew can cause local models to optimise towards different objectives. When these updates are directly averaged using Federated Averaging (FedAvg), the resulting global model may favour categories that are dominant on particular clients. Moreover, overall accuracy may conceal serious degradation in less represented classes. Macro-class accuracy and per-class accuracy are therefore necessary in addition to overall test accuracy.

Federated Proximal (FedProx) introduces a proximal regularisation term into each client’s local objective, limiting excessive deviation from the current global model. It may therefore reduce client drift under strong non-independent and identically distributed (Non-IID) conditions. Nevertheless, its effect can only be interpreted clearly when model initialisation, data partitions, optimisers, learning rates, and training budgets are strictly controlled.

This project uses the German Traffic Sign Recognition Benchmark (GTSRB) to construct a reproducible five-client federated image-classification pipeline. Dirichlet distributions are used to simulate label heterogeneity ranging from mild to extreme, and FedAvg and FedProx are evaluated through both overall and class-level performance. The study aims not only to compare final accuracy, but also to identify the practical robustness boundary of FedAvg and determine when FedProx provides meaningful, although potentially limited, mitigation. Preliminary experiments have already established the technical feasibility of this research design.

2 Preliminary Work and Research Gap

The preliminary work has established a complete experimental pipeline covering GTSRB validation, image preprocessing, fixed client partitioning, local training, sample-weighted server aggregation, model evaluation, and checkpoint recovery. Centralised, Local-only, and IID FedAvg baselines have also been implemented. All main experiments use a convolutional neural network without Batch Normalisation, avoiding interference from incorrectly maintained non-trainable Batch Normalisation statistics during federated aggregation and model export.

Under the IID partition, FedAvg achieved a test accuracy of 92.60% and a macro-class accuracy of 89.44%. It substantially outperformed the independently trained Local-only

models and achieved performance close to the centralised baseline. These results demonstrate that federated aggregation can compensate for the limited amount of training data available to an individual client.

The project subsequently created fixed Dirichlet partitions with α values of 10, 1.0, 0.5, and 0.1. FedAvg remained close to the IID baseline under mild and moderate label skew. At $\alpha=0.5$, the reduction in overall accuracy was small, whereas macro-class accuracy showed a more visible decline. Under the extreme $\alpha=0.1$ setting, test accuracy fell to 71.49% and macro-class accuracy to 69.42%, with several classes receiving zero accuracy. These results suggest that FedAvg does not fail simply because the data are Non-IID. Instead, it remains relatively robust until label heterogeneity becomes sufficiently severe, at which point both overall performance and class-level reliability deteriorate.

Strictly paired preliminary FedProx experiments were also completed. With the same initial model, client partitions, optimiser, server update, and training budget, increasing the proximal strength from $\mu=0$ to $\mu=0.1$ improved test accuracy under $\alpha=0.1$ from 77.51% to 84.52% and macro-class accuracy from 73.60% to 80.79%. Improvements were also observed under $\alpha=0.5$. Most classes benefited, although several remained weak or declined. The evidence therefore supports partial mitigation by FedProx rather than a complete solution to Non-IID learning.

Several research gaps remain. The formal experiments use one fixed random seed and one saved partition, so variation caused by model initialisation and alternative client allocations has not yet been quantified. The current setting also includes only five clients, full participation in every round, and one local epoch, producing less client drift and sampling noise than many practical federated systems. In addition, the present study concentrates on label heterogeneity and does not model differences in lighting, weather, camera quality, viewpoint, or image blur. Multi-seed experiments, partial participation, additional local epochs, and feature-distribution heterogeneity therefore remain important directions for future validation.

3 Objectives and Research Questions

The overall objective is to develop a reproducible federated traffic sign recognition framework, systematically analyse the effect of label heterogeneity on FedAvg, and evaluate whether FedProx can reduce performance degradation under strong Non-IID conditions.

The specific objectives are to:

1. establish Centralised, Local-only, and FedAvg baselines and determine whether federated aggregation compensates for limited client-level data;
2. construct label skew ranging from mild to extreme using different Dirichlet α values and measure changes in overall, macro-class, and per-class performance;
3. compare FedProx with $\mu=0$ and $\mu=0.1$ under strictly matched experimental conditions;
4. evaluate convergence, class-level failure, and estimated communication cost alongside predictive accuracy.

The study addresses three research questions:

RQ1: Can FedAvg achieve performance close to centralised training and higher than Local-only training under IID conditions?

RQ2: At what degree of label heterogeneity do the overall and class-level performance of FedAvg begin to deteriorate substantially?

RQ3: Under strictly matched conditions, can the proximal regularisation used by FedProx mitigate the degradation caused by strong label skew?

4 Methods

4.1 Research Design

The study uses a simulated five-client federated-learning setting. With the dataset, model architecture, client partitions, and evaluation procedure fixed, the experiments first establish Centralised, Local-only, and IID FedAvg baselines. They then examine FedAvg under different levels of Dirichlet label skew and finally compare a FedAvg-equivalent control with FedProx under strong heterogeneity. Only one primary experimental factor is changed at each stage to preserve interpretability. The overall five-client federated-learning procedure is

illustrated in Figure 1.

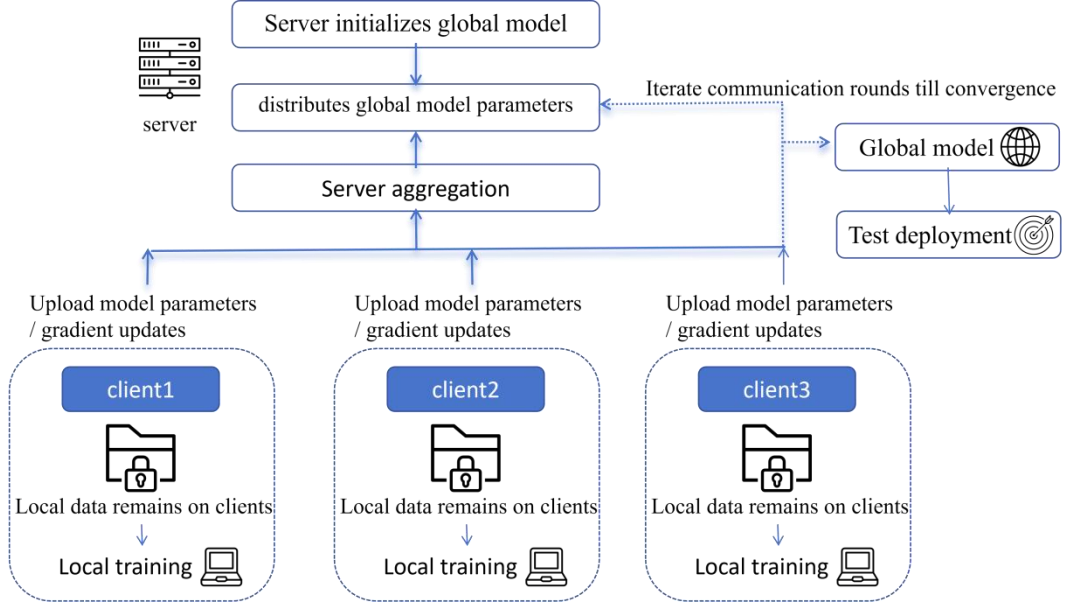


Figure 1. Proposed five-client federated-learning workflow.

4.2 Dataset and Preprocessing

GTSRB contains 39,209 training images, 12,630 official test images, and 43 traffic sign classes. Images are converted to RGB, resized to 32×32 , cast to float32, and normalised to $[0,1]$. Training datasets use shuffling, batching, and prefetching. Horizontal flipping is excluded because it may change the meaning of direction-sensitive signs. The official test set is not used for partitioning, model training, or hyperparameter selection and is evaluated only after the experimental configuration has been fixed.

4.3 Client Partitions

The IID baseline uses an approximately balanced, class-stratified partition in which all five clients contain all 43 classes. Non-IID partitions are generated by sampling a symmetric Dirichlet distribution for each class:

$$P_c \sim \text{Dirichlet}(\alpha, \dots, \alpha)$$

The study uses $\alpha=10, 1.0, 0.5$, and 0.1 to represent progressively stronger label skew. A smaller α concentrates each class on fewer clients. All partitions are generated with the fixed seed 20260720 and stored as client CSV files. Every algorithm reuses exactly the same saved partitions.

4.4 Model and Baselines

The common model is a convolutional neural network without Batch Normalisation:



Figure 2. Common CNN architecture for the proposed federated experiments.

The network outputs 43 logits and uses sparse categorical cross-entropy with sparse categorical accuracy. Batch Normalisation is excluded because earlier federated experiments showed that its moving statistics were not correctly maintained in the previous state-management procedure, producing unreliable global inference.

Centralised training uses the complete training dataset. In the Local-only setting, each client trains independently without parameter exchange. Weighted FedAvg distributes the current global model to all clients, performs local training, and aggregates the resulting updates according to the number of training examples on each client.

4.5 FedAvg and FedProx

The main FedAvg experiments use full participation by all five clients, one local epoch per round, a batch size of 64, and 160 communication rounds. Across the Dirichlet conditions, only the client data distribution changes.

FedProx modifies the local objective as follows:

$$\mathcal{L}_k(w) = \mathcal{L}_{task,k}(w) + \frac{\mu}{2} ||w - w^t||_2^2$$

where w^t is the global model at the beginning of a communication round and μ controls the penalty applied to local deviation. Formal paired experiments compare $\mu=0$ and $\mu=0.1$ under $\alpha=0.1$ and $\alpha=0.5$. Both conditions use identical initial weights, data partitions, client ordering, client-side SGD, learning rate, server update, and training budget. The observed difference can therefore be attributed primarily to proximal regularisation.

4.6 Evaluation and Reproducibility

The best checkpoint is selected according to accuracy on the combined validation set. After parameter selection is complete, the selected model is evaluated on the official test set.

The main metrics are test accuracy, test loss, macro-class accuracy, per-class accuracy, confusion matrix, best communication round, and estimated communication volume. Macro-class accuracy assigns equal importance to all 43 classes, while per-class accuracy and confusion matrices identify whether degradation is concentrated in particular classes or class pairs.

For reproducibility, the project stores fixed client partitions, configuration files, round-level metrics, best and final models, checkpoints, per-class results, and confusion matrices. Paired experiments also verify identical initial weights, finite model updates, sample-weighted aggregation, and continuity after checkpoint recovery. Because the formal results are based mainly on one random seed, conclusions will be restricted to the current model, client population, and training configuration.

5 Expected Outcomes

The project is expected to produce a reproducible five-client federated traffic sign recognition pipeline that supports consistent comparison among Centralised, Local-only, FedAvg, and FedProx training.

The study is expected to identify the robustness boundary of FedAvg under increasing label heterogeneity. Mild and moderate label skew may produce limited changes, whereas extreme skew is expected to reduce overall accuracy, macro-class accuracy, and the performance of specific traffic sign categories. The analysis should also demonstrate why overall accuracy alone is insufficient for identifying class-level failure.

FedProx is expected to provide partial improvement under strong label skew, but not to eliminate all Non-IID-related problems. Its predictive benefits will be considered together with convergence behaviour and communication cost, and all conclusions will be limited to the experimental conditions tested.

The final research outputs will include fixed data partitions, unified training scripts, checkpoints, class-level results, and a reproducible evaluation procedure. These assets will support subsequent studies involving multiple random seeds, partial client participation, additional local epochs, other federated optimisation methods, and feature-distribution heterogeneity.

6 Detailed Timeline

As this proposal is being prepared during the research period, the timeline combines the original research plan with the actual progress achieved to date. The Summer Research Workshop runs from 3 July to 31 July 2026, during which the research proposal, final report, and online presentation are to be completed.

Period	Main Activities	Target Milestone
3-10 July 2026	Conduct the literature review, configure the experimental environment, validate GTSRB, and establish the client data pipeline	Research scope and reproducible experimental setup completed
11-17 July 2026	Implement and evaluate the Centralised, Local-only, and IID FedAvg baselines	Baseline models and evaluation results completed
18-23 July 2026	Construct fixed Dirichlet Non-IID partitions and analyse FedAvg under different levels of label skew	Non-IID FedAvg experiments and preliminary findings completed
24-27 July 2026	Implement FedProx, select parameters using validation results, and complete paired experiments under strong label skew	FedProx results and class-level analysis completed
28-30 July 2026	Consolidate the experimental findings and prepare the Research Proposal, Final Report, and presentation	Submission-ready documents and presentation prepared

31 July 2026	Deliver the online presentation and complete the Workshop requirements	Research project and Workshop presentation completed
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7 Literature and References

7.1 Literature Review

Federated Averaging, introduced by McMahan et al., provides the fundamental federated-learning mechanism used in this project [1]. FedAvg allows clients to perform local optimisation before their model updates are aggregated by a central server. It therefore enables collaborative model training without requiring the transfer of raw client data. Its simple structure and relatively low communication complexity make it a suitable baseline for federated image classification. However, its performance may deteriorate when local client data follow substantially different distributions.

FedProx, proposed by Li et al., extends FedAvg by adding a proximal regularisation term to the local training objective [2]. This term penalises excessive deviation of a local model from the current global model and is intended to improve training stability under statistical and systems heterogeneity. Because FedProx retains a server aggregation procedure similar to FedAvg, it is suitable for controlled experiments in which the proximal term is the main changed variable.

The traffic sign classification experiments use the German Traffic Sign Recognition Benchmark introduced by Stallkamp et al. [3]. GTSRB contains 43 traffic sign categories collected from real road environments and is widely used as a benchmark for multi-class traffic sign recognition.

Hsu et al. examined the effect of non-identical client data distributions on federated visual classification [4]. Their work demonstrates that federated-learning performance can be highly sensitive to the degree and form of data heterogeneity. It also provides methodological support for using parameterised client partitions, including Dirichlet-based label distributions,

to generate controllable Non-IID conditions.

The survey by Kairouz et al. provides a broader overview of federated learning, including optimisation, privacy, communication efficiency, statistical heterogeneity, and systems challenges [5]. It highlights that realistic federated environments rarely satisfy the independent and identically distributed assumption. Together, these studies provide the theoretical and methodological foundations for the dataset selection, federated baselines, Non-IID partition design, and optimisation methods used in this project.

7.2 references

- [1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, PMLR, vol. 54, pp. 1273 – 1282, 2017.
- [2] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” Proceedings of Machine Learning and Systems, vol. 2, pp. 429 – 450, 2020.
- [3] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, “The German Traffic Sign Recognition Benchmark: A multi-class classification competition,” in 2011 International Joint Conference on Neural Networks, pp. 1453 – 1460, 2011, doi: 10.1109/IJCNN.2011.6033395.
- [4] T.-M. H. Hsu, H. Qi, and M. Brown, “Measuring the effects of non-identical data distribution for federated visual classification,” arXiv preprint arXiv:1909.06335, 2019.
- [5] P. Kairouz, H. B. McMahan, et al., “Advances and open problems in federated learning,” Foundations and Trends in Machine Learning, vol. 14, nos. 1 – 2, pp. 1 – 210, 2021, doi: 10.1561/22000000083.

8 Budget

Total requested: HKD 0. This project uses the publicly available GTSRB dataset, open-source TensorFlow and TensorFlow Federated software, and existing personal computing resources. No additional hardware, paid datasets, cloud-computing services, or experimental consumables are required for completion of the proposed work. No reimbursable project expenditure is anticipated, and no research funding is requested.

9 Recommendation

I am pleased to recommend Ma He for the successful completion of the Summer Research Workshop 2026. During the programme, Ma He demonstrated strong initiative, perseverance, and a good capacity for independent technical learning while conducting the research project entitled “Federated Traffic Sign Recognition in Vehicular Networks under Non-IID Data Distribution.”

Throughout the project, Ma He developed a federated learning framework for traffic sign recognition using the GTSRB dataset and TensorFlow Federated. The work included the establishment of centralized, local-only, and FedAvg baselines, the construction of reproducible Non-IID client data distributions, and a systematic investigation of model performance under different degrees of label distribution skew. Ma He also extended the study through controlled FedProx experiments to examine whether proximal regularisation could improve model performance under more severe data heterogeneity.

Ma He showed good technical judgement in identifying and analysing implementation issues encountered during the project. In particular, the investigation of Batch Normalization in the federated training process demonstrated an ability to move beyond surface-level experimental results and consider the underlying model state and aggregation mechanisms. Appropriate adjustments were subsequently made to establish a more reliable and reproducible experimental baseline.

The project addresses a practically relevant and academically meaningful problem at the intersection of federated learning, computer vision, and intelligent transportation. The current study provides useful observations on the robustness and limitations of federated optimisation under heterogeneous client data. It also establishes a sound foundation for further research into advanced federated algorithms, feature and scenario heterogeneity, communication efficiency, and applications in more realistic vehicular environments. Although the project was conducted within the limited timeframe of the Workshop, it has clear potential for continued and more in-depth investigation.

In addition to technical implementation, Ma He demonstrated an appropriate awareness of experimental fairness, controlled comparison, reproducibility, and the limitations of conclusions based on a single experimental setting. The project documentation, analysis, and

presentation of results reflect a careful and responsible approach to research.

Overall, Ma He has made good use of the Summer Research Workshop to develop valuable skills in independent research, experimental design, critical analysis, and academic communication. I consider the work to be thoughtful, technically well organised, and well aligned with the objectives of the programme. I therefore recommend Ma He's project for successful completion of the Summer Research Workshop 2026.